Fingerprints of data: hashing

Cyber security · Level 4 · Lesson 1



In this lesson you will

- Describe the three rules of a good hash.
- Explain how websites store passwords as fingerprints.
- Use fingerprints to check that a file wasn’t changed.

Key ideas

- Any data in, a fixed-size fingerprint out
- Same input, same fingerprint; one letter, a new one
- One-way: no going back
- Salt and slow hashes slow attackers down

Activity: Fingerprint detective

1. Warm-up (3 min): The password puzzle
2. Lab (6 min): Make fingerprints
3. Detect (5 min): Which file changed?
4. Explain (3 min): Tell a partner

Work in pairs with one phone or computer. Before you start, each of you writes two short sentences on paper.

Think it through

1. Why does a good website never need to know your password?
2. What would go wrong if a hash could be run backwards?
3. Why does a long passphrase still matter if the site uses hashes?

Before you go

- One use of a hash, and why it works.